Limits – 9

Sequences defined by integrals and sums

1 сентября 2026 года

Theoretical minimum.

Theorem 1 (Riemann sums). If f is integrable on $[0,1]$, then $\frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right) \to \int_0^1 f$; for monotone f the error is at most $\frac{|f(1)-f(0)|}{n}$, and for smooth f it equals $\frac{f(1)-f(0)}{2n} + O(n^{-2})$.

Theorem 2 (oscillation and averaging). If g is periodic with period T and integrable, then $\int_a^b f(x)g(nx)\,dx \to \frac{1}{T}\int_0^T g \cdot \int_a^b f$; in particular $\int_a^b f(x)\sin(nx)\,dx \to 0$ (the Riemann–Lebesgue lemma).

Fact 1. If the mass of the integral concentrates near the point of maximum, Laplace's method applies: $\int_0^1 f(x)^n dx \sim C\,\frac{f(x_0)^n}{\sqrt{n}}$, where x_0 is the point of maximum; the simplest instance is $\int_0^{\pi/2} \sin^n x\,dx \sim \sqrt{\frac{\pi}{2n}}$.

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1. Prove Theorem 1 together with the error estimate, and compute $\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} \frac{1}{1+(k/n)^2}$ and $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{n}{n^2+k^2}$.

2. Prove Theorem 2 first for a step function f and then for an integrable one; deduce the Riemann–Lebesgue lemma and compute $\lim_{n\to\infty} \int_0^{\pi} \frac{\sin x}{1+3\cos^2(2nx)}\,dx$.

3. Let $a_n = \int_0^1 \frac{x^n}{1+x}\,dx$. Find the asymptotics of a_n, prove that $\sum_{n\geqslant 1}(-1)^n a_n$ converges and compute its sum.

4. (IMC 2025 shortlist) Let $f\colon [0,+\infty) \to [0,+\infty)$ be continuous with $f(x) \to C > 0$ as $x \to +\infty$. For every positive integer n put $a_n = \int_{n^2}^{n^2+1} \big(f(t)\big)^n dt$ and $b_n = \int_{(n-1)^2}^{(n-1)^2+1} \big(f(t)\big)^n dt$. Suppose that $a_1 + a_2 + \cdots + a_n \to \infty$ and $\frac{a_{n+1}}{a_1+a_2+\cdots+a_n} \to L$. Find $\lim_{n\to\infty} \frac{b_1+b_2+\cdots+b_n}{a_1+a_2+\cdots+a_n}$ in terms of C and L.

5. Prove by Laplace's method that $\int_0^{\pi/2} \sin^n x \, dx \sim \sqrt{\frac{\pi}{2n}}$ and deduce Wallis' formula; compare with the exact answer through the gamma function.